# Companion L: The Growth Index in the Relaxation-Driven Cyclic Cosmology

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**Abstract**

The growth index $\gamma$ provides a compact description of the dynamical evolution of cosmic structure. In the standard cosmological model, $\gamma$ is nearly constant, reflecting the scale-independent growth of density perturbations. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the growth rate in a way that renders the growth index scale-dependent and time-dependent. This Companion develops a continuous description of the growth index in RDCC, deriving how the relaxation scale influences the growth rate, the scale dependence of $\gamma$, and the observational signatures in redshift-space distortions, weak lensing, and velocity fields. The resulting framework provides a dynamical fingerprint of the RDCC gravitational field.

## 1 Introduction

The growth index $\gamma$ is defined through the relation

$$f(a) = \Omega_m(a)^\gamma, \tag{1}$$

where $f(a)$ is the linear growth rate. In the standard cosmological model, the growth index is nearly constant, with $\gamma \approx 0.55$ for a Universe dominated by cold dark matter and a cosmological constant.

In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the growth rate in a scale-dependent manner, rendering the growth index both scale-dependent and time-dependent. The growth index becomes a direct probe of the relaxation scale $k_{\rm rel}$ and the temporal structure of the RDCC gravitational field.

## 2 The RDCC Growth Rate

The linear growth rate is defined as

$$f(k,a) = \frac{d \ln D(k,a)}{d \ln a}, \tag{2}$$

where $D(k,a)$ is the growth factor.

In RDCC, the growth factor evolves as

$$D_{\rm RDCC}(k,a) = D(a)\left[1 - \chi_D \left(\frac{k}{k_{\rm rel}}\right)^\alpha\right]. \tag{3}$$

The growth rate becomes

$$f_{\rm RDCC}(k,a) = f(a)\left[1 - \chi_f \left(\frac{k}{k_{\rm rel}}\right)^\alpha\right]. \tag{4}$$

This modification introduces a scale dependence that reflects the relaxation mechanism.

## 3 The RDCC Growth Index

The growth index is defined implicitly by

$$f_{\mathrm{RDCC}}(k,a) = \Omega_m(a)^{\gamma_{\mathrm{RDCC}}(k,a)}. \tag{5}$$

Solving for $\gamma_{\mathrm{RDCC}}$ yields

$$\gamma_{\mathrm{RDCC}}(k,a) = \frac{\ln f(a) + \ln \left[1 - \chi_f \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha}\right]}{\ln \Omega_m(a)}. \tag{6}$$

This expression shows that the growth index becomes scale-dependent and acquires a distinctive RDCC signature near the relaxation scale.

## 4 Scale Dependence and RDCC Signatures

In the standard cosmological model, the growth index is nearly constant. In RDCC, the scale dependence becomes

$$\frac{\partial \gamma_{\mathrm{RDCC}}}{\partial k} \propto -\chi_f\, \alpha\, \frac{k^{\alpha-1}}{k_{\mathrm{rel}}^{\alpha}}. \tag{7}$$

This modification produces:

- a suppressed growth index at small scales, - an enhanced growth index at large scales, - a characteristic turnover near $k_{\mathrm{rel}}$.

These signatures provide a direct observational probe of the relaxation scale.

## 5 Observational Probes

The growth index can be measured through several observables:

- **Redshift-space distortions (RSD)** measure $f\sigma_8$.
- **Weak lensing** measures the combination $f\Omega_m$.
- **Velocity fields** measure the velocity divergence.
- **Cluster abundance** depends on the growth history.

In RDCC, all of these observables acquire a scale-dependent modification that reflects the relaxation mechanism. The growth index therefore provides a dynamical fingerprint of the RDCC gravitational field.

## 6 Conclusion

The growth index in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the growth rate in a scale-dependent manner, rendering the growth index both scale-dependent and time-dependent. These effects produce distinctive signatures in redshift-space distortions, weak lensing, and velocity fields. The present Companion establishes the theoretical foundation for future studies of cosmic growth, dynamical evolution, and the interplay between matter and gravity in RDCC.

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